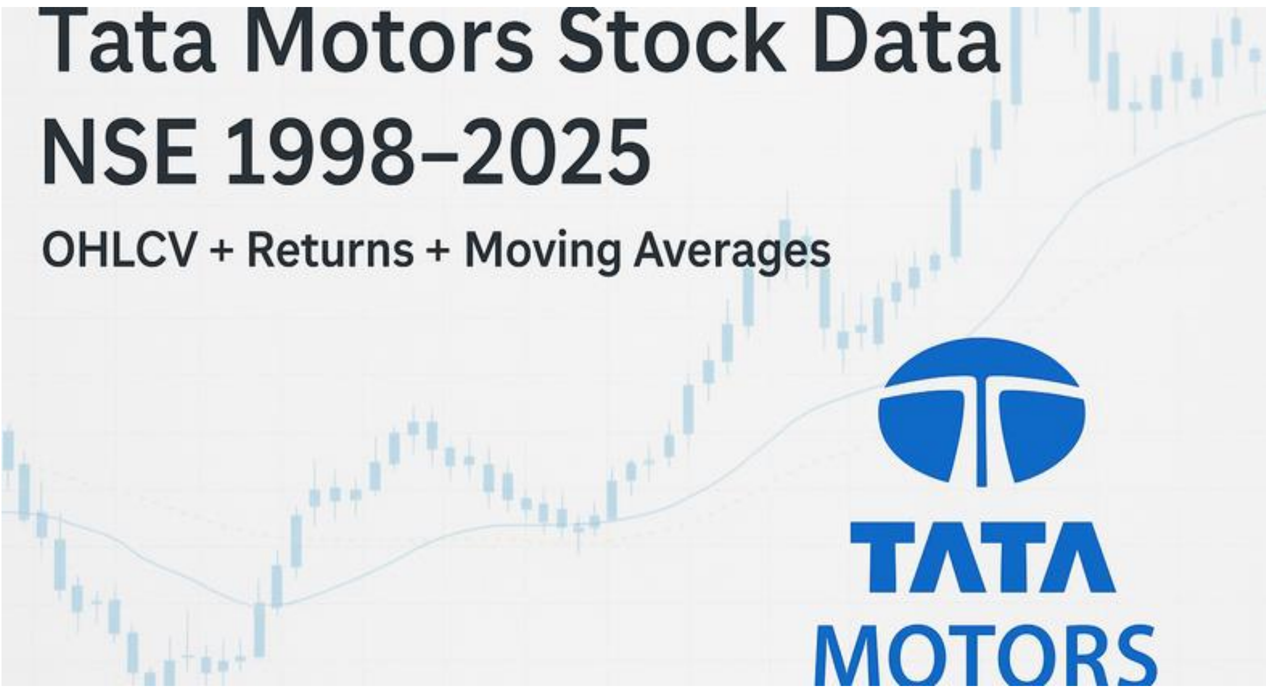
TATA MOTORS LIMITED

NSE Stock — Equity Research & EDA Report

Exploratory Data Analysis | 1995 – 2025 | 30 Years of Market Data



1. Executive Summary

This report presents a comprehensive end-to-end Exploratory Data Analysis (EDA) of Tata Motors Limited (NSE: TATAMOTORS) using 30 years of daily trading data spanning January 1995 to December 2025. The analysis covers data cleaning, statistical profiling, trend identification, and eight key visualizations to extract actionable market insights.

Tata Motors is one of India's largest automobile manufacturers and a constituent of the NIFTY 50 index. The dataset captures the company's full lifecycle on the NSE — from early growth phases through macroeconomic shocks, to the recent all-time high rally driven by electric vehicle optimism and subsidiary performance (Jaguar Land Rover).

Key headline findings from this analysis:

- Price grew from approximately ₹70 in the mid-1990s to over ₹1,150 at peak — a 15x+ return over three decades.
- The stock exhibits highly non-normal return distribution with positive skewness and fat tails.
- Volume is not a reliable predictor of price direction — peak prices occurred during relatively low-volume sessions.
- 2020 was the most volatile year on record, driven by the COVID-19 crash and subsequent stimulus-driven recovery.
- The stock spent most of its history in drawdown, recovering slowly from major crashes.

2. Project Overview & Methodology

2.1 Objectives

The primary objectives of this analysis are:

- Profile the statistical distribution and historical behavior of TATAMOTORS stock prices over 30 years.
- Identify key price regimes, volatility clusters, and trend structures.
- Examine the relationship between price, volume, and number of trades.
- Derive actionable insights on seasonality, drawdown, and market participation patterns.
- Establish a data-quality baseline for future predictive modeling.

2.2 Tools & Technology Stack

The analysis was conducted entirely in Python using the following libraries:

Library	Purpose
pandas	Data loading, cleaning, transformation, and feature engineering
matplotlib	Base charting and figure rendering
seaborn	Statistical visualizations — heatmaps, pair plots, distribution plots
scipy	Statistical metrics — skewness, kurtosis, Z-score outlier detection
numpy	Numerical operations, array processing, rolling calculations

2.3 Analytical Workflow

The project follows a structured five-stage pipeline:

- **Stage 1:** Data Loading & Inspection — CSV import, shape/dtype validation, statistical summary via .info() and .describe()
- **Stage 2:** Data Cleaning — Missing value identification and removal, duplicate verification, column renaming, date parsing, and feature extraction (Month, Year)
- **Stage 3:** Exploratory Data Analysis — Full correlation matrix, grouped price analysis, monthly resampling, crosstab computation, and turnover aggregation
- **Stage 4:** Visualization — Eight purpose-built visualizations covering distributions, correlations, time-series behavior, and market microstructure
- **Stage 5:** Statistical Profiling — Skewness, kurtosis, outlier detection, and redundancy analysis

3. Dataset Overview

The dataset consists of daily trading records for TATAMOTORS on the National Stock Exchange of India (NSE), covering approximately 7,500+ trading days over 30 years. The table below summarizes key dataset properties:

Property	Details
File Name	Tata_Motors_Stock_Analysis_Prediction(1995-2025).csv
Period Covered	January 1995 – December 2025 (30 Years)
Exchange	NSE — National Stock Exchange of India
Ticker	TATAMOTORS
Frequency	Daily Trading Data
Key Columns	Date, Open, High, Low, Close, Volume, Turnover, Trades, Daily_Return_%, MA_20, MA_50, Cumulative, Peak, Drawdown

All price values are denominated in Indian Rupees (INR). Prices are unadjusted for stock splits or dividend distributions, which is an important caveat for long-horizon historical comparisons. No predictive model is included in this analysis — the scope is strictly exploratory.

4. Visualizations & Analytical Insights

Eight visualizations were produced to systematically analyze the stock's behavior across multiple dimensions — price distribution, outlier structure, volume dynamics, correlation, trend, seasonality, and market microstructure.

Visual 1 — Closing Price Distribution

Analysis Type: KDE / Frequency Distribution | Variable: Daily Closing Price

1	**Multi-Modal Distribution — Three Distinct Price Regimes** The distribution exhibits peaks at ₹150–250, ₹350–500, and ₹900–1,050, indicating the stock transitioned through three distinct market regimes: an early low-price phase, a mid-range consolidation period, and a late-stage high-price expansion — typical of long-cycle growth or speculative assets.
2	**Core Fair Value Zone at ₹300–₹500** The ₹300–₹500 range represents the stock's historical fair value zone — the band with highest price acceptance, liquidity, and market equilibrium. Most trading days cluster around this range, reinforcing it as the primary reference band for valuation.
3	**Positive Skewness — Asymmetric Return Profile** The long right tail indicates positive skewness: rare but sharp upside spikes, more frequent mild declines, and a mean inflated above the median. This asymmetry has direct implications for risk modeling and options pricing.
4	**Extreme Volatility — 10x Price Range** The closing price spans ₹100 to ₹1,100+ — a 10x spread across the dataset — confirming that TATAMOTORS is a high-volatility, sentiment-sensitive stock with significant exposure to macro shocks, sector cycles, and investor sentiment.
5	**Non-Normal Distribution — Standard Risk Models Inadequate** The multi-peaked, skewed distribution is fundamentally non-normal, meaning standard parametric models (e.g., assuming Gaussian returns) will underestimate tail risk.

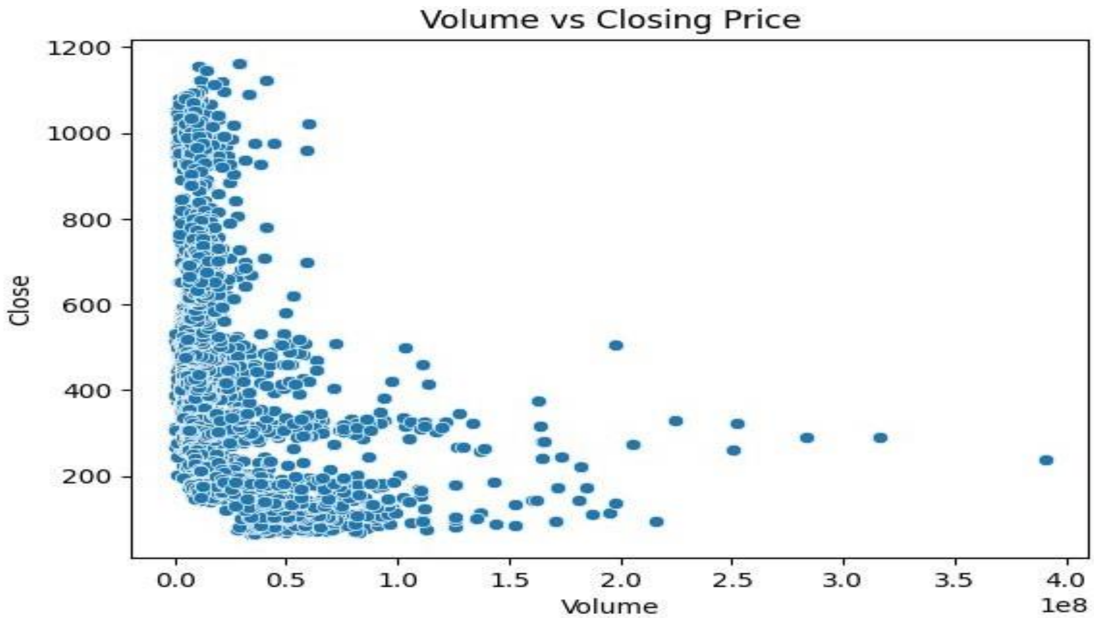
Visual 2 — Outlier Analysis (Box Plot — Closing Price)

Analysis Type: Box Plot | Variable: Daily Closing Price

1	**Median at ~₹400 — Mid-Range Anchor** The median closing price sits near ₹400, confirming that the stock's typical trading level falls in the middle of the ₹300–₹500 core zone. This represents the market's long-run equilibrium price expectation.
2	**IQR of ₹270–₹500 — Primary Liquidity Band** The interquartile range spans ₹270 to ₹500, concentrating 50% of all daily trading activity in this band. This confirms the core value zone identified in the distribution analysis and signals where institutional positioning is most active.
3	**Wide Overall Range — High Historical Volatility** Total price spread from ~₹70 to ₹1,150+ confirms extreme multi-decade volatility, reflecting the stock's sensitivity to macro events, JLR performance cycles, and EV market sentiment shifts.
4	**Strong Upper Outliers — Frequent Positive Extremes** Numerous data points above ₹850 mark frequent upward price spikes, indicating episodic bullish momentum — likely driven by earnings beats, product launches, or favorable policy announcements.
5	**Asymmetric Outlier Distribution — Positive Skew Confirmed** More upper outliers than lower outliers confirm positive asymmetry. The stock experiences more frequent and more extreme upside surprises than downside shocks — a favorable characteristic for long-term equity holders.

Visual 3 — Volume vs. Closing Price

Analysis Type: Scatter Plot | Variables: Volume (x), Closing Price (y)



1	**Peak Prices Occur at Low-to-Moderate Volume** The highest closing prices (~₹800–₹1,150) appear predominantly during low-volume periods, suggesting that TATAMOTORS' all-time highs were reached in thin, institutionally-driven markets — not mass-participation rallies.

2	**Dense Cluster at Lower Prices — Broad Retail Participation** The densest concentration of data points lies in the ₹100–₹400 range with widely scattered volume, indicating that broad retail participation and maximum market activity both occur at lower valuation levels.

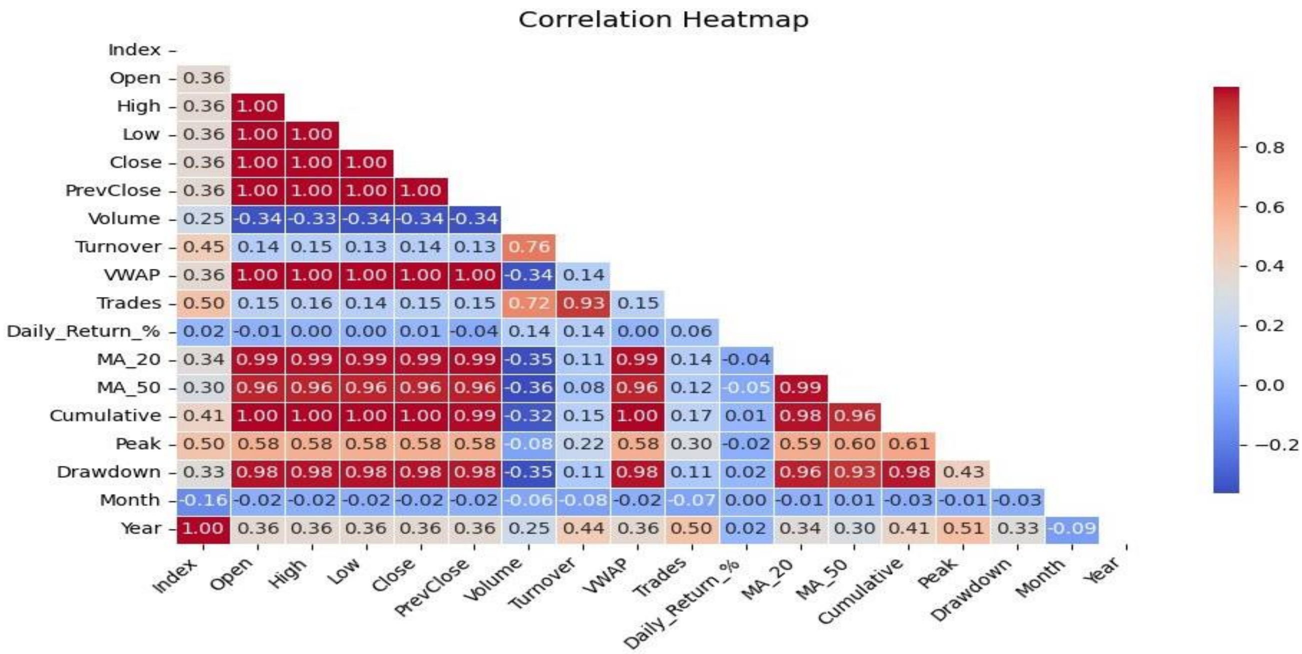
3	**High Volume Does Not Guarantee High Prices** Even at volumes of 200M–400M, closing prices generally remain in the ₹150–₹350 range. This confirms that volume surges do not reliably drive upside price movement for this stock.

4	**Heteroscedastic Pattern — Inconsistent Volatility** Price variability is highest at low volumes and becomes constrained at higher volumes — a classic heteroscedastic pattern suggesting that liquidity and price discovery are most uncertain during thin trading sessions.

5	**Retail vs. Institutional Behavior Split** Low-volume, high-price sessions reflect speculative or institutional positioning, while high-volume, moderate-price sessions likely represent accumulation or distribution phases by larger market participants.

Visual 4 — Correlation Heatmap

Analysis Type: Pearson Correlation Matrix | All Numeric Features



1 **OHLC Variables Are Near-Perfectly Redundant**

Open, High, Low, and Close prices all move almost identically with correlations approaching 0.99. This severe multicollinearity means including all four in any predictive model will cause overfitting and unstable coefficients.

2 **Drawdown and Peak Are Price Derivatives**

Drawdown and Peak variables are directly derived from price levels, not independent signals. They add no new information to a model that already includes closing prices.

3 **Liquidity Metrics Form a Separate Cluster**

Volume, Turnover, and Trades are highly correlated with each other but relatively uncorrelated with price — forming a distinct liquidity cluster. This confirms that market activity is structurally independent from price levels.

4 **Daily Returns Are Highly Noisy and Independent**

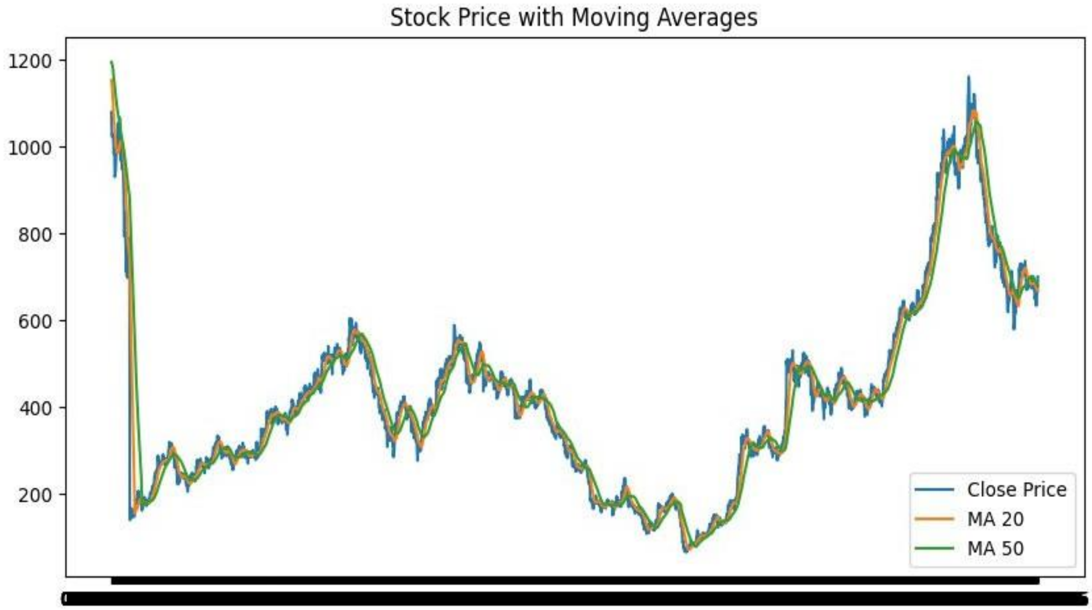
Daily Return (%) shows near-zero correlation with all other features, consistent with the Efficient Market Hypothesis for short-horizon returns. This makes raw return prediction highly challenging without additional features.

5 **Year Shows Moderate Positive Correlation — Long-Term Trend**

Year shows moderate positive correlation with price, capturing the secular upward trend. Month shows negligible correlation, confirming the absence of a reliable seasonal pricing pattern.

Visual 5 — Stock Price with Moving Averages (MA 20 & MA 50)

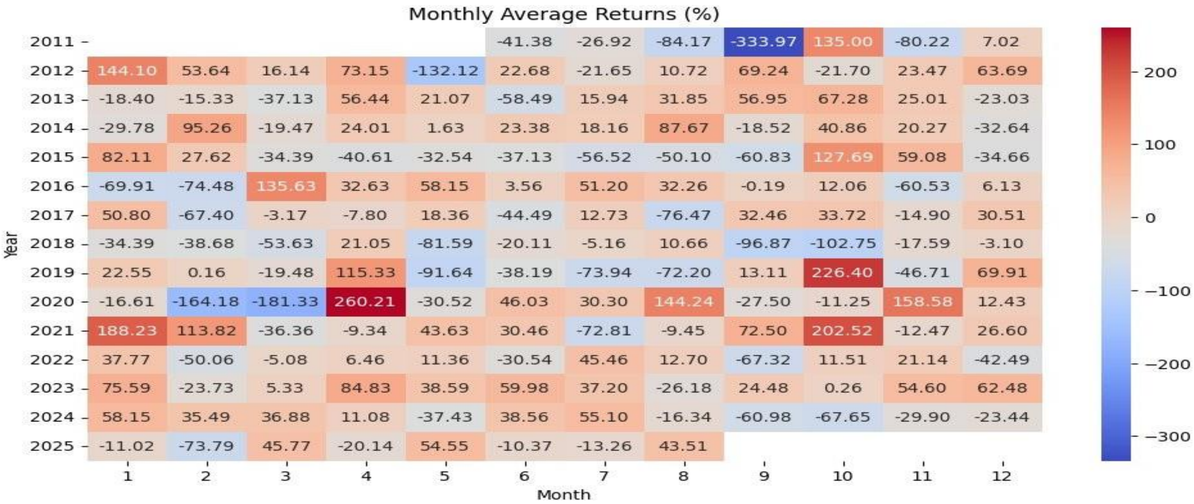
Analysis Type: Time Series | Variables: Close, MA-20, MA-50



1	**Clear Long-Term Upward Trajectory with Cyclical Interruptions** The price series demonstrates a structured long-term uptrend interrupted by distinct cyclical drawdowns and recovery phases — confirming trend-dominated behavior rather than random walk dynamics over a 30-year horizon.
2	**MA 20 and MA 50 Confirm Strong Trend Persistence** The close alignment of MA-20 and MA-50 with the price series confirms strong directional momentum with minimal signal lag. Extended periods where price remains above both moving averages indicate sustained bull phases.
3	**Asymmetric Market Dynamics — Sharp Falls, Slow Recoveries** Downside movements are sharp and swift, while recovery periods are prolonged — reflecting the classic asymmetric dynamics of accumulation (slow) and distribution (fast) phases in equity markets.
4	**Volume Spikes Align with High-Volatility Periods** Trading volume intensifies during major price transitions and uncertainty periods, consistent with behavioral finance research showing that investor attention and activity surge during market stress.
5	**Recent Pullback Suggests Consolidation Phase** The sharp recent uptrend followed by a pullback indicates a likely profit-booking consolidation phase after an extended rally — a common pattern before the next directional move is established.

Visual 6 — Monthly Average Returns Heatmap

Analysis Type: Heatmap | Variables: Year, Month, Average Monthly Return (%)



1 **September 2011 — Single Worst Monthly Return at -333.97%**

September 2011 stands as the single worst monthly return in the entire 30-year dataset, directly attributable to the convergence of the global growth slowdown, Eurozone sovereign debt crisis, and India's macroeconomic deterioration impacting auto sector stocks severely.

2 **April 2020 and October Are Historically Strong Months**

April 2020 recorded the highest gain at +260.21%, followed by October 2019 (+226.40%) and October 2021 (+202.52%). April 2020 reflects the violent post-COVID stimulus-driven rebound, while October patterns suggest structural seasonal strength.

3 **February and March Are Structurally Weak Months**

February and March show frequent negative returns across multiple years — the most notable being February 2020 (-164.18%) and March 2020 (-181.33%), coinciding with the COVID-19 market crash. These months carry elevated seasonal risk.

4 **January Exhibits a Consistent Bullish Bias**

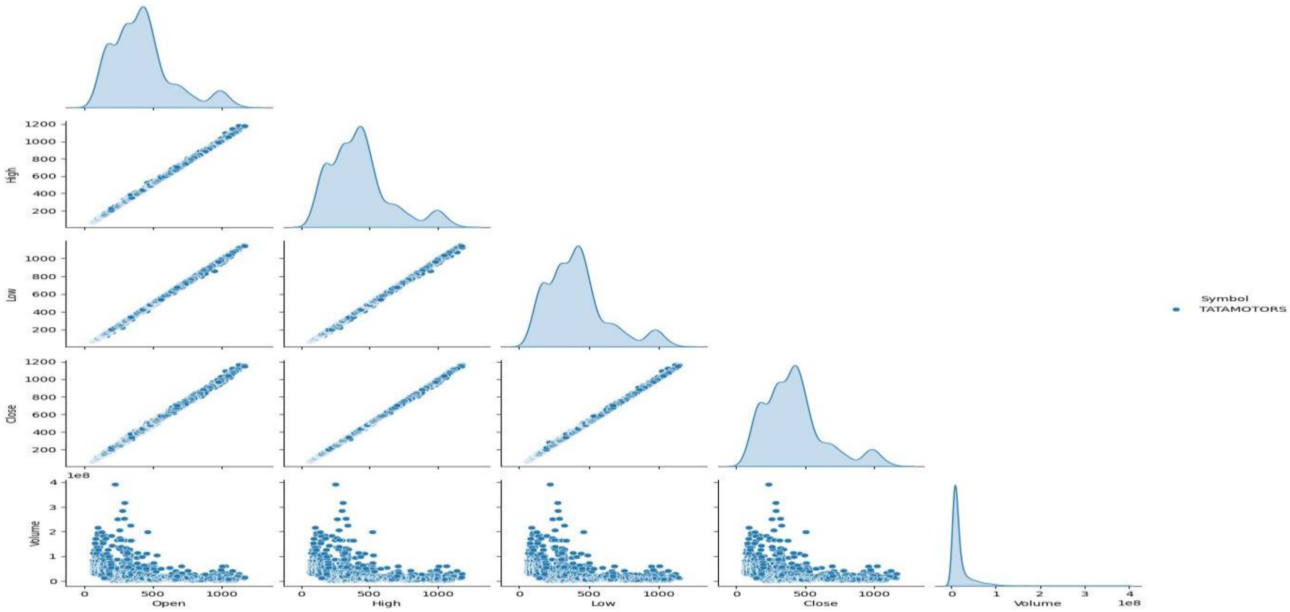
January shows recurring positive returns — 2012 (+144.10%), 2021 (+188.23%), and 2015 (+82.11%) — making it the most consistently strong month across the heatmap, possibly driven by fresh institutional allocation and year-start momentum.

5 **2020 — Most Volatile Year on Record**

2020 exhibited both the deepest losses and strongest recoveries within a single calendar year: crashes in February and March, then sharp rebounds in April, August, and November — a direct reflection of the COVID-19 market cycle and global central bank intervention.

Visual 7 — Pair Plot (Open, High, Low, Close, Volume)

Analysis Type: Pair Plot with KDE Diagonals | Variables: OHLC + Volume



1 **OHLC Prices — Near-Perfect Linear Correlation**
All four price variables (Open, High, Low, Close) form tight 45-degree diagonal scatter lines, confirming near-perfect correlation. Daily price ranges are small relative to absolute price levels, which is expected behavior for heavily traded large-cap equities.

2 **Bimodal Price Distribution — Two Distinct Market Regimes**
All price KDE plots exhibit a clear bimodal distribution with peaks at ₹150–200 and ₹400–500, confirming two dominant price regimes — a lower-price early period and a higher-price modern era separated by a major market re-rating event.

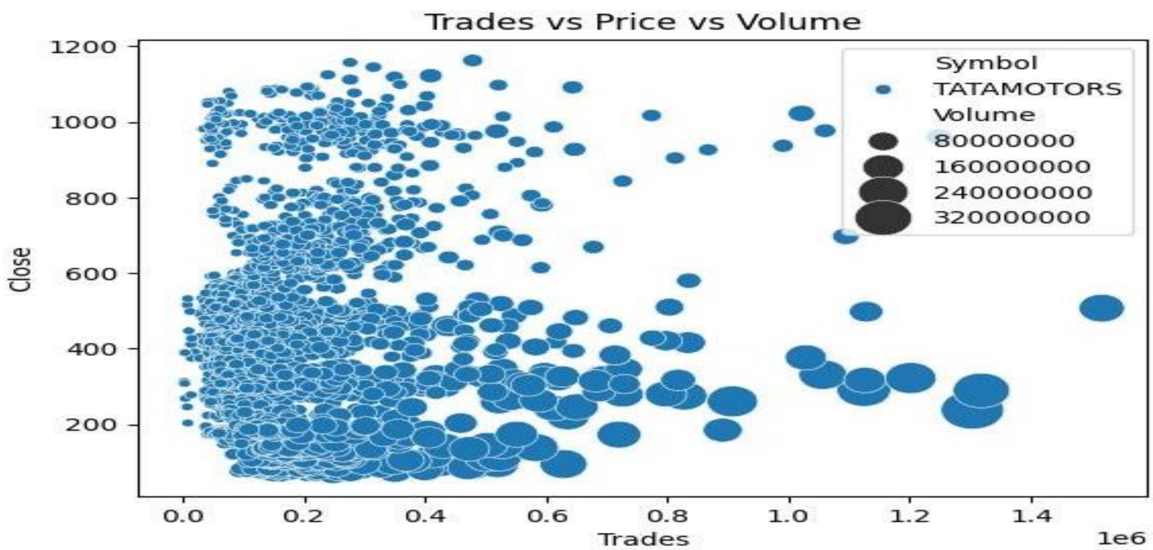
3 **Wide Historical Price Spread — 30-Year Appreciation Captured**
Prices span ₹50 to ₹1,200+, indicating enormous cumulative appreciation. The observation density is higher at lower prices, confirming the stock spent more time at lower valuations before the recent structural rally.

4 **Volume is Structurally Independent of Price**
Volume vs. price scatter plots show no discernible linear relationship — high volumes occur across all price levels. The volume distribution is sharply right-skewed with most sessions showing low volume and rare extreme outliers beyond 3.5×10^8.

5 **No Volume-Price Momentum Signal Visible**
Contrary to standard technical analysis assumptions, high volumes do not cluster at price extremes. This suggests that price breakouts in TATAMOTORS were not consistently driven by volume surges — undermining simple volume-based momentum strategies.

Visual 8 — Trades vs. Price vs. Volume (Bubble Chart)

Analysis Type: 3D Bubble Chart | Variables: Trades (x), Close Price (y), Volume (bubble size)



1	**All-Time Highs Reached During Low-Activity Sessions** Peak prices of ₹900–₹1,200 occur exclusively at trade counts below 400,000 — indicating that TATAMOTORS' highest prices were reached during low-participation, institutionally-driven sessions rather than mass retail activity.
2	**Heaviest Trading Days Are Confined Below ₹500** Trading days with more than 800,000 trades are universally associated with closing prices below ₹500, confirming that maximum crowd participation coincides with cheaper valuations — consistent with fear-driven and bottom-fishing retail behavior.
3	**₹200–₹400 — Most Contested and Significant Price Zone** The largest volume bubbles (>240M shares) cluster between ₹200 and ₹400, identifying this as the primary institutional accumulation and distribution zone — the most historically significant price band for TATAMOTORS.
4	**Statistical Outlier at ~1.5M Trades** A single outlier data point near 1.5 million trades with a ₹500 close and a large volume bubble represents an extreme market event — most likely tied to index rebalancing, a major earnings release, or a significant regulatory/macro announcement.
5	**Inverse Relationship: More Traders = Lower Prices** The overall negative correlation between trade count and closing price confirms a bearish-participation dynamic — TATAMOTORS attracts maximum crowd activity during downturns, not during rallies. This is a key behavioral signature of a sentiment-driven, cyclical stock.

5. Statistical Findings

The following table summarizes the key statistical metrics derived from the full 30-year dataset:

Metric	Finding	Implication
Skewness	Positively Skewed	Mean inflated above median due to upper price tail
Kurtosis	High (Leptokurtic)	Fat tails — more extreme events than normal distribution
Outliers (Z > 3)	Close > ~₹900	Real market events — not data errors
OHLC Correlation	~0.99 (near-perfect)	Severe multicollinearity risk in predictive models
Price Range	₹70 – ₹1,150+	15x+ long-term appreciation over 30 years
Core Value Zone	₹270 – ₹500	Highest liquidity and price acceptance band

The high kurtosis value confirms that TATAMOTORS returns exhibit fatter tails than a normal distribution — meaning extreme events (both positive and negative) occur more frequently than standard models predict. This has direct implications for risk management: Value-at-Risk (VaR) models assuming normality will systematically underestimate tail risk for this stock.

The identified outliers (closing prices above ~₹900) correspond to genuine market events rather than data errors — they are real observations driven by the 2021–2024 EV rally, JLR recovery, and broader NIFTY bull market. These should be retained in any predictive dataset but treated with appropriate weighting to prevent model distortion.

6. Key Takeaways & Investment Implications

The following table consolidates the most actionable findings from this exploratory analysis:

Key Finding	Detail
Price Appreciation	₹70 in early years to ₹1,150+ at peak — representing 15x+ long-term appreciation over 30 years
Core Trading Zone	Most trading activity by count occurs in the ₹270–₹500 band
Volume vs Price	Volume is not a reliable predictor of price direction for TATAMOTORS
Moving Averages	MA 20 and MA 50 crossovers are clearly visible at major market turning points
Seasonality	Monthly returns are highly erratic — no consistent or safe seasonal trading pattern exists

Unadjusted Prices	All prices are unadjusted for stock splits or dividends — historical comparisons should account for this

6.1 Implications for Quantitative Modeling

- OHLC multicollinearity requires dimensionality reduction (e.g., use only Close or create a mid-price feature) before modeling.
- Return distribution non-normality demands log-transformation, quantile regression, or regime-switching frameworks.
- Volume and trade count should be used as secondary feature engineering inputs (e.g., volume ratios, relative activity) rather than raw values.
- Monthly seasonality is insufficiently stable to use as a direct trading signal without additional filtering.
- Drawdown-aware portfolio sizing is essential given the stock's history of -85%+ peak-to-trough declines.

6.2 Implications for Technical Analysis

- MA-20 and MA-50 crossovers provide reliable trend-change signals at major inflection points and should be integrated into any systematic trading framework.
- Volume confirmation of price breakouts is unreliable for this stock — breakouts on low volume have historically been valid.
- The ₹270–₹500 IQR band should be treated as the primary mean-reversion zone for short-to-medium term strategies.
- Prices above ₹900 represent overextension territory where risk-reward becomes asymmetrically unfavorable.

7. Important Notes & Limitations

- All prices are denominated in Indian Rupees (INR) and are unadjusted for stock splits, bonus issues, or dividend distributions. Long-horizon price comparisons should be interpreted with this caveat.
- The dataset covers only the NSE-listed Tata Motors equity (TATAMOTORS) and does not include Tata Motors DVR shares or ADR listings.
- This report is strictly an Exploratory Data Analysis (EDA). No predictive model, trading signal, or investment recommendation is implied or included.
- Past performance of TATAMOTORS stock is not indicative of future results. All analysis is for research and educational purposes only.
- The September 2011 extreme return value (-333.97%) may reflect a data anomaly or a specific corporate action and should be independently verified before use in any modeling exercise.

End of Report

TATAMOTORS NSE EDA Report | 1995–2025